

RHYTHM DISORDERS AND ELECTROPHYSIOLOGY

CLINICAL CASE

Concealed Wenckebach Phenomenon in Left Bundle Branch Associated With Chest Pain



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ABSTRACT

BACKGROUND Painful left bundle branch block (LBBB) syndrome is a rare cause of chest pain associated with intermittent LBBB.

CASE SUMMARY We report the case of a 68-year-old man who experienced chest pain during a treadmill test. The temporal relationship between chest pain and the concealed Wenckebach phenomenon in the left bundle branch, along with the absence of documented myocardial ischemia, strongly suggested a diagnosis of painful LBBB syndrome. The patient underwent up-titration of a β -blocker.

DISCUSSION The chest pain in painful LBBB syndrome is thought to be related to dyssynchrony in myocardial contraction caused by intermittent depolarization disturbances. Although rare, the Wenckebach phenomenon can complicate conduction in the bundle branches, leading to intermittent LBBB. (JACC Case Rep. 2025;30:103751)

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HISTORY OF PRESENTATION

A 68-year-old man with a history of hypertension, dyslipidemia, and chronic coronary syndrome underwent treadmill testing for interscapular pain radiating anteriorly and associated with exertion. The test was stopped during the second minute of the second stage of the Bruce protocol due to the beginning of pain that replicated his reported symptoms.

The onset of pain coincided with the onset of intermittent left bundle branch block (LBBB) on the

TAKE-HOME MESSAGES

- The Wenckebach phenomenon is not limited to the atrioventricular node; when it involves the bundle branches, it can lead to intermittent bundle branch block.
- Painful LBBB syndrome is a rare cause of chest pain; it should be considered in the differential diagnosis of patients with exertional angina, including those with a known history of coronary artery disease.

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**ABBREVIATIONS
AND ACRONYMS****BBB** = bundle branch block**ECG** = electrocardiogram**LBB** = left bundle branch**LBBB** = left bundle branch
block

electrocardiogram (ECG) (Figure 1). At that time, his blood pressure was 185/65 mm Hg, and his heart rate was 136 beats/min. In the recovery phase, the intermittent LBBB disappeared, and the chest pain resolved simultaneously. Analysis of the narrow QRS complexes observed after normalization of intraventricular conduction showed no evidence of myocardial ischemia (Figure 2). The echocardiogram performed at the end of the study showed no global or segmental wall motion abnormalities.

PAST MEDICAL HISTORY

One month before, the patient had undergone coronary angiography for exertional angina that revealed an 80% stenosis in the midsegment of the left anterior descending artery, which was treated with angioplasty and drug-eluting stent placement.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis included new onset fixed coronary lesions, early stent failure, coronary spasm, and painful LBBB syndrome.

INVESTIGATIONS

The patient underwent another coronary angiography, which revealed patency of the previously

implanted stent and no new hemodynamically significant lesions (Figure 3). An electrophysiological study was not performed because the patient did not provide his consent.

MANAGEMENT

The patient underwent medical therapy optimization with β -blocker up-titration (from 1.25 mg of bisoprolol to 3.75 mg).

DISCUSSION

The ECG in the Figure 1 (leads V_1 , V_5 , and V_6 recorded simultaneously) shows a regular sinus tachycardia. Most ventricular complexes exhibit a LBBB morphology. However, this pattern normalizes cyclically and reproducibly, with the LBBB disappearing intermittently.

The normalization of intraventricular conduction does not depend on variations in the R-R intervals: cycles ending with narrow complexes are the same as those ending with wide complexes. A similar behavior is observed in the PR intervals. Therefore, it can be concluded that the bundle branch block (BBB) is not heart rate dependent. The disappearance of the LBBB occurs in nonpremature beats, thus ruling out supranormal conduction.

The underlying mechanism of intermittent conductive disorder is represented by concealed

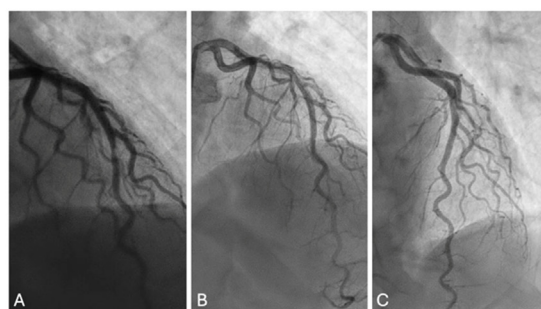
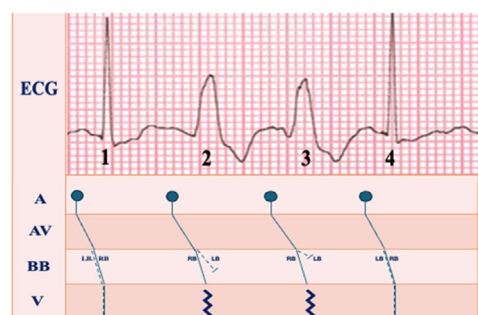
VISUAL SUMMARY Concealed Wenckebach in the Left Bundle Branch and Exertional Chest Pain

FIGURE 1 Electrocardiogram at the Onset of Chest Pain



Leads V₁, V₅, and V₆ recorded simultaneously show regular sinus tachycardia at 136 beats/min with intermittent left bundle branch block, attributable to a concealed Wenckebach phenomenon in the left bundle branch with 3:2 numerical ratio.

Wenckebach phenomenon in the left bundle branch (LBB), with a predominant 3:2 numerical ratio. The Wenckebach phenomenon involves a gradual lengthening of the conduction time for successive impulses until one is blocked. The impulse after the block is conducted better than the one preceding it, and this cycle repeats.

The Wenckebach phenomenon is ubiquitous throughout the conduction tissue and is not confined to the atrioventricular node.¹ Friedberg and Schamroth² were the first to describe the concealed Wenckebach phenomenon in the LBB. The Wenckebach phenomenon in the His-Purkinje system has recently been documented also by endocavitary electrophysiological studies.³

When the Wenckebach phenomenon complicates conduction in a bundle branch, 2 types of periodicities can occur.

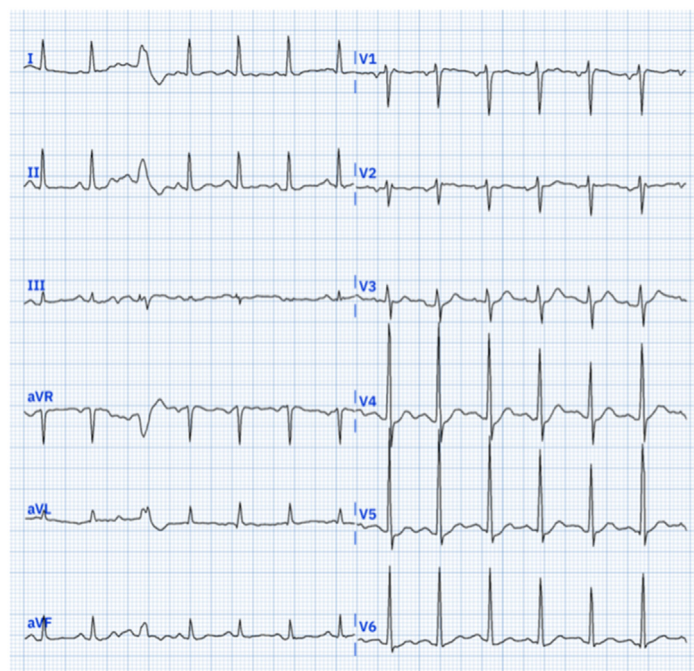
1. Manifest: characterized by progressive increase in intraventricular conduction time until complete BBB occurs.
2. Concealed: can only be deduced from the periodical, unexpected normalization of intraventricular conduction, every 2 or more consecutive QRS displaying complete BBB, unrelated to changes in cycle length.

To better understand the mechanism of BBB concealment in relation to the ECG presentation, it is important to consider that the degree of bundle branch conduction disturbance depends on the conduction delay. The smallest delay that allows us to observe changes in the QRS complex is 0.01 seconds; a delay between 0.04 and 0.06 seconds results in a complete BBB.⁴ It has been hypothesized that such a delay allows the impulse to fully activate the

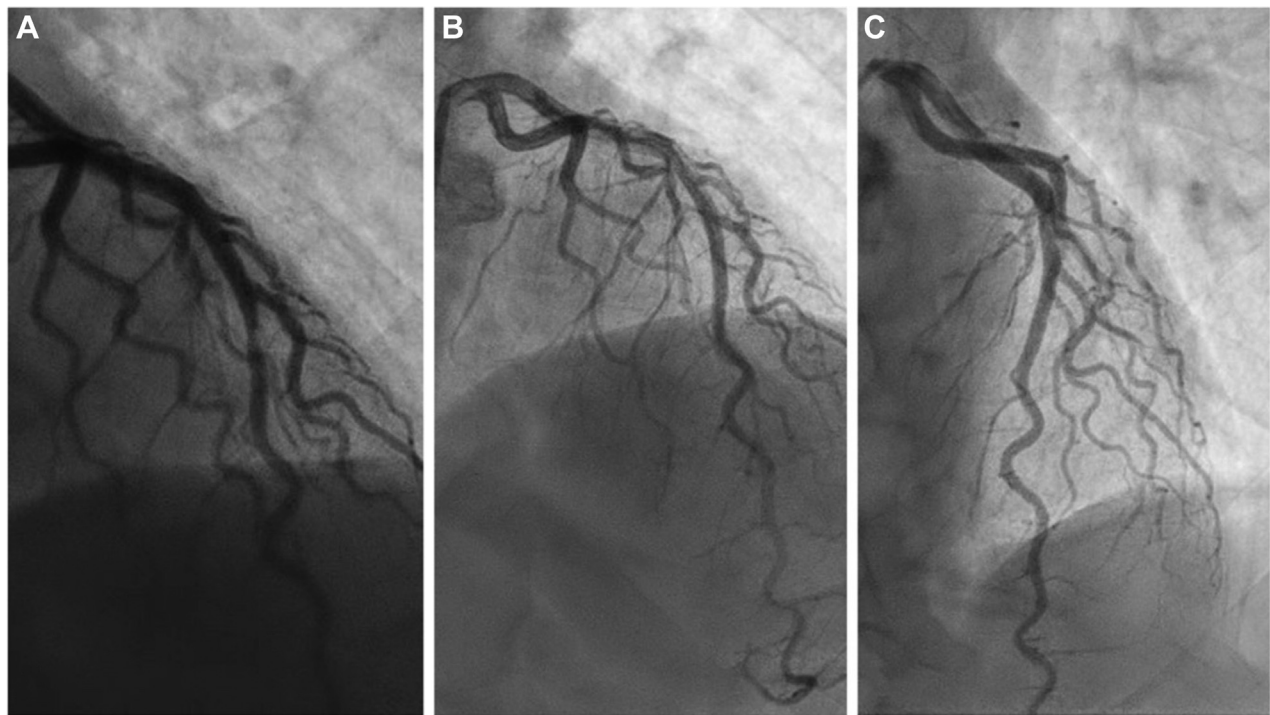
ventricle through the contralateral branch before the wavefront travels anterogradely through the affected branch. Further delaying the conduction time in this branch will not increase the degree of BBB, which will already be at its maximum.

The ladder diagram (Figure 4) provides a detailed explanation of the phenomenon. The first impulse of

FIGURE 2 Electrocardiogram After Normalization of Intraventricular Conduction



The electrocardiogram shows the normalization of intraventricular conduction with regular sinus tachycardia at 115 beats/min and no signs of acute myocardial ischemia.

FIGURE 3 Angiographic Finding

Angiographic view of left anterior descending coronary artery showing a tight stenosis in the middle portion of the vessel (A) treated with a drug-eluting stent implantation (B). Coronary angiography was repeated 1 month later, demonstrating stent patency and no new stenosis (C).

the Wenckebach series is conducted simultaneously by both branches, producing a narrow QRS. The second impulse, as expected in a typical Wenckebach sequence, undergoes the maximum delay. The delay is such that the ventricle is activated entirely by the impulse conducted over the right branch, and the resulting QRS displays a pattern of complete LBBB. The third impulse results in a complete conduction failure at the origin of the LBB. However, electrocardiographically, the ventricular complex is indistinguishable from the previous one. After the blocked impulse 3, the LBB has time enough to recover excitability; consequently, the next impulse (labeled 4) is conducted simultaneously by both bundle branches again, indicating the presence of concealed Wenckebach phenomenon indirectly.

Although the most probable diagnosis, considering the clinical history, is exertional angina, our case presents elements that call this diagnosis into question. The chest pain appears suddenly and coincides with the onset of intermittent LBBB, and its

resolution coincides with the disappearance of the LBBB. After the resolution of chest pain and LBBB, no signs suggestive of myocardial ischemia were observed in the analysis of the narrow QRS complexes. Furthermore, the coronary angiography showed good results from the previously implanted stent with no new hemodynamically significant lesions or new onset fixed coronary lesions.

Another explanation for the characteristic electrocardiographic findings and symptoms is coronary vasospasm. This condition is often investigated as a potential cause of unexplained chest pain and is typically ruled out with ergotamine infusion during coronary angiography.⁵⁻⁷ In our case, a spontaneous vasospasm occurred in the proximal segment of the left anterior descending artery during the angiography, which was effectively treated with nitroglycerin. As a result, we decided against administering ergotamine after excluding new-onset fixed coronary lesions and early stent failure. During the vasospasm, the patient did not experience chest pain, and continuous electrocardiographic

monitoring showed no changes in the duration of the ventricular depolarization phase consistent with LBBB. Therefore, it is highly unlikely that the vasospasm was related to the clinical presentation in our case.

The onset and offset of chest pain do not show the typical characteristics of ischemic pain, which generally begins gradually and increases in intensity over several minutes. Ischemic pain typically subsides gradually with rest or after taking medications like nitroglycerin. By contrast, the pain in our patient's case did not follow this gradual onset and resolution pattern, suggesting that it may not have been related to ischemia.

This pattern, especially the temporal relationship between chest pain and LBBB, is highly suggestive of painful LBBB syndrome.⁸ This is a rare diagnosis of chest pain caused by intermittent LBBB in the absence of myocardial ischemia. Dyssynchrony in myocardial contraction has been hypothesized as the main cause of the pain.^{9,10} As has been previously described, painful LBBB syndrome can coexist with coronary artery disease. However, it is often underdiagnosed in this context.⁷

FOLLOW-UP

The patient returned for examination 1 month later and reported no chest pain.

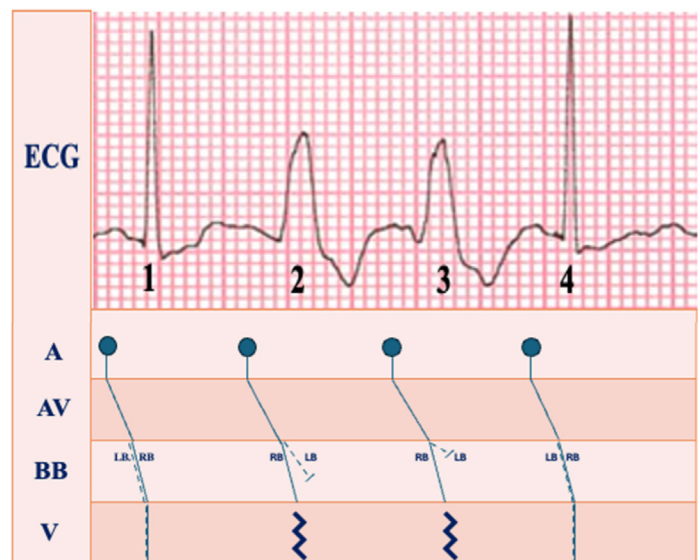
CONCLUSIONS

To the best of our knowledge, we report the first case of painful LBBB syndrome induced by a concealed Wenckebach phenomenon in the LBB. Painful LBBB is a rare cause of chest pain and is characterized by the temporal relationship between chest pain and LBBB. It requires differential diagnosis that includes fixed coronary lesions, coronary spasm, and, in our case, early stent failure. The key criterion for diagnosing painful LBBB syndrome is not the absence of angiographically documented coronary artery disease but rather the absence of objectively documented myocardial ischemia.

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FIGURE 4 Ladder Diagram of Concealed Wenckebach Phenomenon



The ladder diagram represents the atrium, the atrioventricular junction, the bundle branches, and the ventricle. It illustrates a concealed Wenckebach sequence in the left bundle branch (impulses numbered from 1 to 3); the fourth impulse marks the start of a new sequence. Conduction in the right bundle branch is represented by a solid line, and conduction in the left bundle branch is depicted by a dashed line. The symbol > illustrates the delayed conduction resulting from the activation of entire ventricular mass by the contralateral bundle branch. A = atrium; AV = atrioventricular junction; BB = bundle branches; ECG = electrocardiography; LBB = left bundle branch; RBB = right bundle branch; V = ventricle.

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Dr Natale is a consultant for Biosense Webster, Stereotaxis, and Abbott Medical, and has received speaker honoraria/travel from Medtronic, Atricure, Biotronik, and Janssen. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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KEY WORDS exertional angina, intermittent conduction abnormalities, painful left bundle branch syndrome